



1. Compute $\frac{1}{3} + \frac{1}{8} + \frac{1}{15} + \frac{1}{24} + \frac{1}{35}$ in simplest form.

- (A) $\frac{4}{7}$ (B) $\frac{25}{42}$ (C) $\frac{13}{21}$ (D) $\frac{27}{42}$ (E) $\frac{2}{3}$

2. James and Cal start at the same point on a linear track that is 12 miles long. James runs at a constant 4 mph while Cal runs at a constant 6 mph. At the end of the track there is a wall; when Cal reaches the wall, he turns around and runs back toward the start. After how many hours will Cal and James meet?

- (A) 2 (B) $\frac{7}{3}$ (C) $\frac{12}{5}$ (D) $\frac{11}{4}$ (E) 3

3. Janet rolls two fair dice labeled from 1 to 6. What is the probability that the sum on the top faces is a multiple of 3 or 4?

- (A) $\frac{7}{18}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{19}{36}$ (E) $\frac{5}{9}$

4. Mike's class weights: tests 40%, projects 25%, quizzes 20%, homework 15%. Mike has homework 100%, tests 90%, quizzes 90%. What is the lowest project grade to ensure an overall grade at least 90%?

- (A) 84 (B) 85 (C) 86 (D) 87 (E) 88

5. A square S_1 of side 4 sits on top of a square S_2 of side 6. The segment connecting the centers is vertical. Another segment connects the midpoint of the top side of S_1 to the bottom-left corner of S_2 and hits the bottom side of S_1 at a point P . What is the distance from the bottom-left corner of S_1 to P ?

- (A) $\frac{3}{5}$ (B) $\frac{13}{20}$ (C) $\frac{7}{10}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

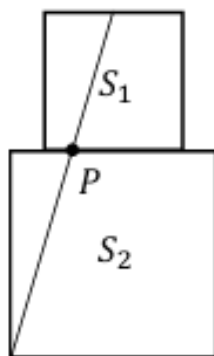


Figure 1: Problem 5

6. A permutation $\{a_1, a_2, a_3, a_4, a_5\}$ of $1, 2, 3, 4, 5$ is called *distant* if $|a_{n+1} - a_n| > 1$ for all $n = 1, 2, 3, 4$. How many distant permutations are there?

- (A) 14 (B) 16 (C) 18 (D) 20 (E) 22

7. Ten teams play a round-robin (each pair meets once; no ties). What is the maximum number of teams that can have an odd number of wins?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

8. Arithmetic sequence A_1 : 30 terms, common difference d , sums to 75. Another sequence A_2 : 75 terms, common difference $-d$, and the first term equals that of A_1 ; all terms of A_2 sum to 30. Find d .

- (A) $\frac{21}{515}$ (B) $\frac{21}{340}$ (C) $\frac{21}{225}$ (D) $\frac{21}{71}$ (E) $\frac{21}{70}$

9. Let S be the set of ordered pairs of natural numbers (a, b) satisfying $\log_{10} a + \log_{10} b = 2016$. Let P be the product of ab over all $(a, b) \in S$. Compute $\log_{10} P$.

- (A) 2016 (B) 2017^2 (C) $2016 \cdot 2017$ (D) $2016 \cdot 2017^2$ (E) $2017!$

10. A regular hexagon has side length 1. What is the least number of distinct points placed on or inside the hexagon to guarantee that some three of them form a nondegenerate triangle of area $< \frac{\sqrt{3}}{16}$?

- (A) 20 (B) 21 (C) 23 (D) 24 (E) 25

11. Zach is trying to catch a train that will arrive at a random time between 2:00 PM and 3:00 PM; whenever the train arrives, it waits 10 minutes before leaving, even if it would leave after 3:00 PM. Zach arrives at a random time between 2:00 PM and 3:00 PM and stays until 3:00 PM. What is the probability that Zach catches the train?

- (A) $\frac{43}{72}$ (B) $\frac{11}{18}$ (C) $\frac{5}{8}$ (D) $\frac{23}{36}$ (E) $\frac{47}{72}$

12. In square $ABCD$ of side length 1, points E, F, G lie on sides AB, BC, CD , respectively. If triangle EFG has a right angle at F , what is the maximum possible area of triangle EFG ?

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{3}$ (C) $\frac{3}{5}$ (D) $\frac{2}{3}$ (E) $\frac{\sqrt{2}}{2}$

13. Define $f_1(x) = 1 + \frac{1}{x}$ and, for integers $n \geq 2$, let $f_n(x) = 1 + \frac{1}{f_{n-1}(x)}$. Which of the following x satisfies $f_{12}(x) = 0$?

- (A) $-\frac{147}{231}$ (B) $-\frac{144}{233}$ (C) $-\frac{89}{144}$ (D) $-\frac{3}{5}$ (E) $-\frac{1}{2}$

14. A cooperative board game proceeds in phases. At the beginning of each phase, the number 2016 is written on the board. In phase k ($1 \leq k \leq 4$), on each turn a player replaces the current number N by the integer $\frac{N}{d_k}$, where d_k is a k -digit positive divisor of 2016 excluding 1 and 2016, and no d_k can be reused within the same phase. When a move would fail to produce an integer, the phase ends and the players advance to phase $k+1$. The last turn occurs in phase 4, where a player erases 1008 and replaces 2016 with 2. If the goal is to complete the game in the least possible total number T of turns, what is T ?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

15. A 3×3 slider puzzle has an open space on the rightmost cell, and the other eight cells each contain a numbered square. A “slide” moves a square adjacent to the open space

3	7	8
4	5	
6	1	2

Figure 2: Problem 15

into that space. After three random slides, what is the probability that the sum of the numbers in the rightmost column is odd?

$$(A) \frac{7}{12} \quad (B) \frac{5}{8} \quad (C) \frac{2}{3} \quad (D) \frac{17}{24} \quad (E) \frac{3}{4}$$

16. Let $D(n)$ denote the number of positive divisors of n . Compute the sum of all positive integers $n < 100$ such that $D(n) + D(n+1) = D(100)$.

$$(A) 15 \quad (B) 49 \quad (C) 64 \quad (D) 130 \quad (E) 145$$

17. Inside a right circular cone C of height 4 and base radius 3, spheres $S_1, S_2, \dots$ are placed as follows: S_1 is tangent to the base and is the largest sphere that fits inside C ; for $n \geq 2$, S_n is externally tangent to S_{n-1} , lies closer to the cone's vertex than S_{n-1} , and is the largest sphere that fits inside C subject to these constraints. Let $V(S_k)$ be the volume of S_k . Compute $\sum_{k=1}^{\infty} V(S_k)$.

$$(A) 4\pi \quad (B) \frac{32\pi}{7} \quad (C) \frac{69\pi}{14} \quad (D) \frac{288\pi}{37} \quad (E) \frac{64\pi}{7}$$

18. Triangle ABC has $BC = 6$, $\angle CAB = 60^\circ$, and $\angle BCA = 80^\circ$. Let F be the foot of the perpendicular from C to the angle bisector of $\angle CAB$, and let M be the midpoint of BC . Find the length MF .

$$(A) \sqrt{2} \sin 20^\circ \quad (B) \sqrt{3} \sin 20^\circ \quad (C) 2 \sin 20^\circ \quad (D) 2\sqrt{3} \sin 20^\circ \quad (E) 3\sqrt{3} \sin 20^\circ$$

19. For $0^\circ \leq x \leq 30^\circ$, the numbers $\cos x$, $\sin x$, and $\cos 2x$ (in some order) form an arithmetic sequence. What is the common difference of this arithmetic sequence?

$$(A) \frac{1}{6} \quad (B) \frac{1}{4} \quad (C) \frac{\sqrt{7}}{8} \quad (D) \frac{\sqrt{2}}{2} \quad (E) \frac{1+\sqrt{7}}{4}$$

20. An equilateral triangle has all three vertices on the same branch of the hyperbola $x^2 - \frac{y^2}{b} = 1$, with one vertex at $(1, 0)$. If its area can be any real number in $[3\sqrt{3}, 12\sqrt{3}]$, then the possible values of b form an interval $[x_1, x_2]$. What is $x_1 + x_2$?

$$(A) \frac{3}{10} \quad (B) \frac{1}{3} \quad (C) \frac{3}{7} \quad (D) \frac{9}{20} \quad (E) \frac{1}{2}$$

21. A palindrome is an integer that reads the same forwards and backwards (with no leading zeroes). How many positive base-10 palindromes less than 2016 are also palindromes in base-6?

(A) 10 (B) 14 (C) 15 (D) 16 (E) 18

22. A monic quadratic $P(x)$ satisfies: (i) all coefficients are positive integers less than 10; (ii) all roots are integers; (iii) exactly two (counted with multiplicity) of the roots of $P(P(x))$ are integers. There exists a monic quartic $Q(x)$ such that the roots r_1, r_2, r_3, r_4 of $Q(Q(x))$ satisfy that the roots of $P(P(x))$ are $r_k - n$ for $1 \leq k \leq 4$ and some nonzero integer n . How many possible polynomials $Q(x)$ exist?

(A) 0 (B) 1 (C) 2 (D) 4 (E) Infinitely Many

23. Define $f : N \rightarrow N$ by $f(1) = 1$ and, for $n \geq 2$,

$$f(n) = \begin{cases} \prod_{i=1}^{n-1} f(i), & \text{if } n \text{ is odd,} \\ \sum_{i=1}^{n-1} f(i), & \text{if } n \text{ is even.} \end{cases}$$

There exists n such that $f(n) = 3^k \prod_{i=0}^{30} \left[2^{i+2} + \sum_{j=0}^{i+1} 2^j f(5 + 2i - 2j) \right]^{30-i}$, for some integer k . What is the remainder when $n + k$ is divided by 1000?

(A) 648 (B) 672 (C) 693 (D) 707 (E) 715

24. Students $A, B, C, D, E, F, G, H, I, J$ stand on the circles in the network shown (with A initially at the topmost point). Each student must move to a neighboring circle so that after moving, no two students occupy the same circle and at most one pair of students swaps positions. How many distinct formations are possible after they move?

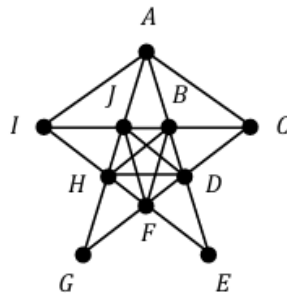


Figure 3: Enter Caption

(A) 22 (B) 52 (C) 62 (D) 70 (E) 94

25. Let $S = \{(x, y, z) : x, y, z \text{ is integer, } |x|, |y|, |z| \leq 1\}$. Some triples of points in S lie on a rotated and translated parabola of the form $y = ax^2 + bx + c$ with $a \neq 0$. How many distinct groups of 4 points in S can exist such that every 3-point subset of the group lies on some such parabola?

- (A) 720 (B) 860 (C) 960 (D) 1080 (E) 1200

Time limit: 75 minutes